

Factor Structure Puzzles

Answer Key

Grades 4–5 Mathematics

Quick Answers

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|--------------------------|--|
| 1. 13 | 8. 7 |
| 2. 7 units | 9. 40 minutes |
| 3. 6 | 10. 16 |
| 4. 12 | 11. (a) the number = 300, — |
| 5. (a) digit sum = 15, — | 12. (a) smallest total = 36, (b) how many of those |
| 6. 30 seconds | smallest totals fit in $540 = 15$, — |
| 7. 7 | |
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What is verified

9 of 12 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 5, 11, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grades 4–5 Mathematics

4.OA.B.4 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12

Difficulty 1–5 of 5 across 16 tagged checks.

Common wrong answers

6. *If they got 60: multiplied the two flash times without dividing out the factor they share*

9. *If they got 160: multiplied the two turn times without dividing out the factor they share*

1. Fill in the missing factor: $7 \times \underline{\hspace{1cm}} = 91$

Solution: Ask which number times 7 gives 91. Count up in sevens, or divide: $91 \div 7 = 13$, and check $7 \times 13 = 91$. 13

2. A rectangle is made from 84 unit squares. One side is 12 units long. How long is the other side?

Solution: The two side lengths are a factor pair of the area, so the missing side is $84 \div 12$. Since $12 \times 7 = 84$, the other side measures 7 units

3. Kai lists every factor pair of 54 and picks the pair whose two numbers are closest together. The larger number of that pair is 9. Write the smaller number of that pair.

Solution: The factor pairs of 54 are 1×54 , 2×27 , 3×18 and 6×9 . The last pair is the closest together, and its partner for 9 is $54 \div 9$. 6

4. A teacher shares 108 stickers into equal packs with none left over. Each pack holds more than ten and fewer than fifteen stickers. How many stickers are in each pack?

Solution: “None left over” means the pack size is a factor of 108. The factors are 1, 2, 3, 4, 6, 9, 12, 18, 27, 36, 54, 108. Only one of them lies between ten and fifteen, and its partner factor is 9, since $108 \div 9 = 12$. 12

5. Ivy wants to know whether 4236 is divisible by 6. (a) Write the sum of the digits 4, 2, 3 and 6. (b) Is the number divisible by 6? Explain using the test for 2 and the test for 3.

Solution (a): $4 + 2 + 3 + 6 = 15$. 15

(b) — instructor-judged. A number is divisible by 6 exactly when it passes both smaller tests. Test for 2: the last digit is 6, which is even, so it passes. Test for 3: the digit sum 15 is a multiple of 3, so it passes. Passing both means the number is divisible by 6, and indeed $4236 = 6 \times 706$.

Look for a response that names *both* tests. A response that checks only the digit sum has shown divisibility by 3, not by 6.

6. A red lamp flashes every 6 seconds and a blue lamp every 10 seconds. They flash together right now. How many seconds pass before they next flash together?

Solution: List the multiples. Red: 6, 12, 18, 24, 30. Blue: 10, 20, 30. The first number on both lists is 30.

The factor structure says the same thing: $6 = 2 \times 3$ and $10 = 2 \times 5$; the shared factor 2 must not be counted twice, so the least common multiple is $6 \times 10 \div 2$. 30 seconds

Common wrong answer: 60 — multiplying the two times together. That is a common multiple, but not the first one.

7. Write 126 as a product of prime numbers, then write its largest prime factor.

Solution: Split repeatedly until only primes remain: $126 = 2 \times 63 = 2 \times 3 \times 21 = 2 \times 3 \times 3 \times 7$.

Dividing the prime factors already found out of 126 leaves the largest one: $126 \div (2 \times 3 \times 3)$. 7

8. In the three-digit number $4?7$, the ? stands for one missing digit. Which digit makes the whole number divisible by 9?

Solution: A number is divisible by 9 when its digit sum is a multiple of 9. The two known digits give $4 + 7 = 11$, so the missing digit must lift the sum to the next multiple of 9, which is 18: the digit is $18 - 11$. 7

Check: 477 has digit sum 18, and $477 = 9 \times 53$.

9. One ferris wheel finishes a turn every 8 minutes and another every 20 minutes. They start together. How many minutes pass before they next start a turn together?

Solution: Multiples of 8: 8, 16, 24, 32, 40. Multiples of 20: 20, 40. The first shared one is 40.

By factor structure, $8 = 2 \times 2 \times 2$ and $20 = 2 \times 2 \times 5$; they share $2 \times 2 = 4$, so the least common multiple is $8 \times 20 \div 4$. 40 minutes

Common wrong answer: 160 — the product of the two times, with the shared factor of 4 counted twice.

10. Mina arranges 96 tiles into a rectangle with equal rows. The number of rows is a factor of 96 that is greater than five and less than eight. How many tiles are in each row?

Solution: The factors of 96 are 1, 2, 3, 4, 6, 8, 12, 16, 24, 32, 48, 96. Between five and eight there is exactly one: 6 rows. Its partner factor is $96 \div 6$. 16

11. The prime factorisation of a mystery number is $2 \times 2 \times 3 \times 5 \times 5$. (a) What is the mystery number? (b) Is it divisible by 15? Explain how the list of prime factors tells you, without dividing.

Solution (a): Multiply the list: $2 \times 2 = 4$, $4 \times 3 = 12$, $12 \times 5 = 60$, $60 \times 5 = 300$. 300

(b) — instructor-judged. Yes. Since $15 = 3 \times 5$, and the list contains a 3 and a 5, those two can be pulled out and multiplied to make 15; what stays behind is $2 \times 2 \times 5$, a whole number. So the mystery number is 15 times a whole number.

Look for a response that points at the 3 and the 5 inside the list. A response that simply divides has answered the question but not the one that was asked.

12. A bakery packs cookies into boxes of 12 or trays of 18. (a) What is the smallest total greater than zero that fills whole boxes and whole trays? (b) The bakery has 540 cookies. How many of those smallest totals fit into 540? (c) Explain how the prime factors of 12 and 18 show that 540 works.

Solution (a): A total that works is a common multiple of 12 and 18. Multiples of 12: 12, 24, 36. Multiples of 18: 18, 36. The first shared one is 36. Using factor structure instead, $12 = 2 \times 2 \times 3$ and $18 = 2 \times 3 \times 3$ share $2 \times 3 = 6$, so the answer is $12 \times 18 \div 6$. 36

(b) Now divide the stock by that smallest total: $540 \div 36$, and $36 \times 15 = 540$. 15

(c) — instructor-judged. Look for a response that builds 36 out of the primes: filling whole boxes needs $2 \times 2 \times 3$ in the total, filling whole trays needs $2 \times 3 \times 3$, and the smallest total carrying both is $2 \times 2 \times 3 \times 3 = 36$. Because 540 is 36 multiplied by a whole number, it carries those primes too, so it fills whole boxes and whole trays. Full credit requires the prime-factor argument, not repeated trial division.